Anjali Vathanakumaran

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Education

University of Waterloo | BASc in Computer Engineering with a Mechatronics Option September 2023 - May 2028
Relevant Coursework: Computer Networks, Real-Time Operating Systems, and Systems Programming and Concurrency

Technical Skills

Languages: Python, C, C++, C#/.Net, HTML, CSS, Java, JavaScript, MATLAB
Frameworks & Libraries: TensorFlow, PyTorch, ONNX, NVIDIA TensorRT, YOLO, CNN-based architectures
Software & Web: Django, RESTful APIs, GraphQL, MySQL, Node.js, React

Work Experience

Firmware Team Lead | *Electrium Mobility, Waterloo, ON* December 2025 – September 2026

- Leading the development of a **BLDC** electronic speed controller (ESC), working on embedded motor control firmware involving state-machine-based control, PWM generation, sensor feedback, and closed-loop control validation.

Computer Vision Engineering Intern | *Martinrea International Inc, Vaughan, ON* May 2026 – August 2026

- Engineered a **React**/**HTML**-based front-end and **C++** backend to bring the standardized **Python**-based vision system into a live inspection application, integrating a **Django-based** data-offloading pipeline to sync edge-device results to a centralized server.
- Designed a human-in-the-loop vision system accommodating real-time inspection for **13 assemblies** to reduce manual inspection time, selecting cameras and hardware while evaluating and integrating **YOLO**, **TensorRT**, and **ONNX** models.

Computer Vision Application Intern | *Martinrea International Inc, Vaughan, ON* September 2025 – December 2025

- Re-architected and standardized a production-grade **Python** codebase for industrial **ML-based computer vision systems** into a **multiprocessing** architecture, adopted as the core codebase for CV deployments across manufacturing plants globally.
- Adapted the vision system for a **conveyor belt production line**, developing a **ghost-box tracking algorithm** to maintain part identity under variable camera frame rates, preventing duplicate or dropped detections that could stall production.
- Developed and trained a **three-stage model pipeline** (**detection**, **classification**, and **segmentation**) using machine learning architectures to perform sequential quality checks on each part, reducing defect detection false negatives to **0.77%**.

Software Developer Intern | *The Honest Consultants, Toronto, ON* January 2025 – April 2025

- Designed and implemented scalable backend services using **Node.js**, building dynamic data synchronization pipelines between cloud services, implementing schema-flexible **GraphQL** queries to support various client data models.
- Developed internal tooling dashboards using **HTML/CSS** and server-rendered templates, automating client configuration workflows and reducing manual operational overhead.

Software Engineering Intern | *Graze AI, Toronto, ON* May 2024 – August 2024

- Built and scaled production-grade web scraping automation systems in **Python**, engineering robust data pipelines that extracted millions of data points and reduced data sourcing costs by over **$10k** annually.
- Architected and iteratively benchmarked multiple scraping pipelines using **Django, Selenium and Scrapy**, selecting optimal approaches to reduce runtime by **60%** and improve reliability through automated data enrichment.

Projects

Real-Time Executive (RTX) Kernel January 2026

- Built a **multitasking kernel** in **C** on the **STM32** (ARM Cortex-M4), implementing task scheduling APIs, TCB-based state management, and interrupt-driven context switching to support safe concurrent execution.

Multi-Core CPU Simulator August 2025

- Designed and implemented a **multi-core CPU simulator** in **C++**, building custom dynamic dequeues and memory management mechanisms to support efficient task scheduling, runtime resizing, and performance-optimized workload execution.

Water Reservoir Monitoring System May 2025 – August 2025

- Designed and built an embedded water reservoir monitoring prototype using an **STM32 microcontroller**, selected sensors, and supporting ICs; developed **C++ firmware** to implement control logic and real-time system behaviour.

Fall Detection Classifier January 2025 - April 2025

- Advanced the development of a **fall detection device** by developing and testing various **binary classification and TensorFlow-based CNN models** in **Python (TensorFlow)** to classify barometric pressure sensor data as fall or no-fall events.